

기초전기실험

13주차

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13 주차 실험 목표

■ AC Ch. 10 : Series-parallel sinusoidal circuits

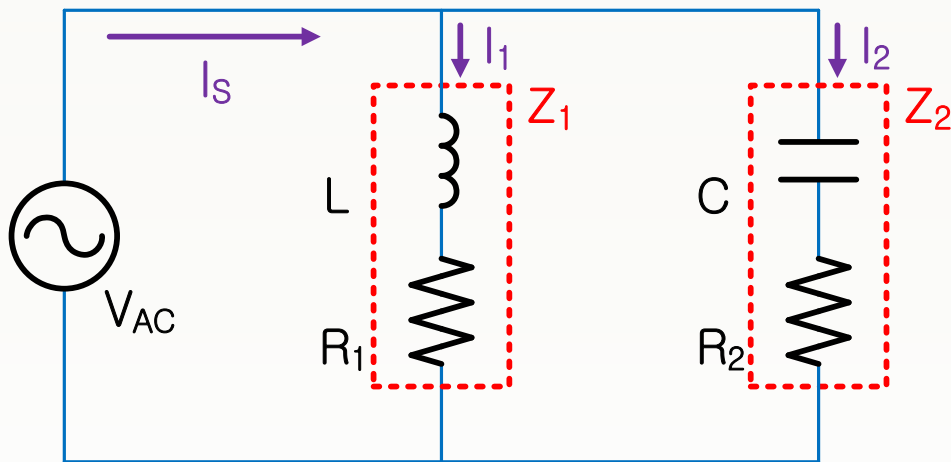
- 직렬 및 병렬로 연결된 R-L, R-C, R-L-C 회로에 교류 전원을 가할 때, Kirchhoff's voltage law, Kirchhoff's current law가 성립함을 확인한다.
- 각 소자에 걸리는 전압과 회로에 흐르는 전류를 측정하고, 위상차를 구한다.

■ AC Ch. 12 : Thevenin's theorem and maximum power transfer

- 실험을 통하여 Thevenin's theorem이 성립함을 확인한다.
- Thevenin's theorem이 DC회로와 AC회로에 적용되는 경우의 차이를 알아낸다.
- Maximum power가 전달되는 저항 값을 계산해 보고, 실험을 통하여 확인한다.

SERIES-PARALLEL SINUSOIDAL CIRCUITS

■ Impedance in a Series-Parallel Circuit



By Ohm's law

$$I_s = \frac{V_{AC}}{Z_T}$$

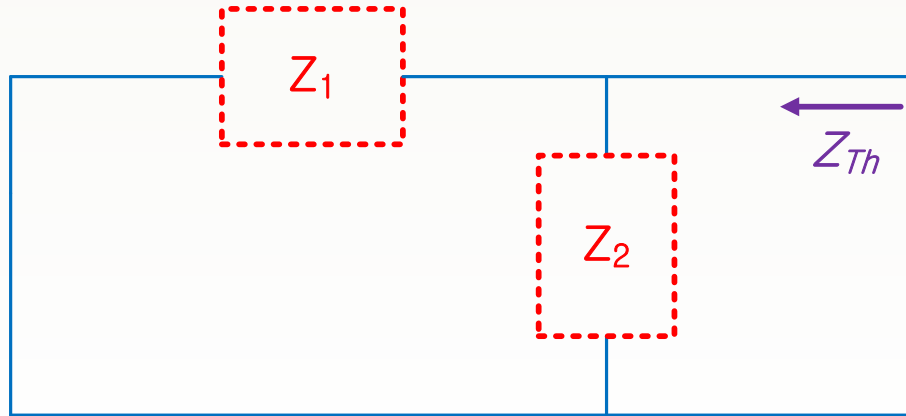
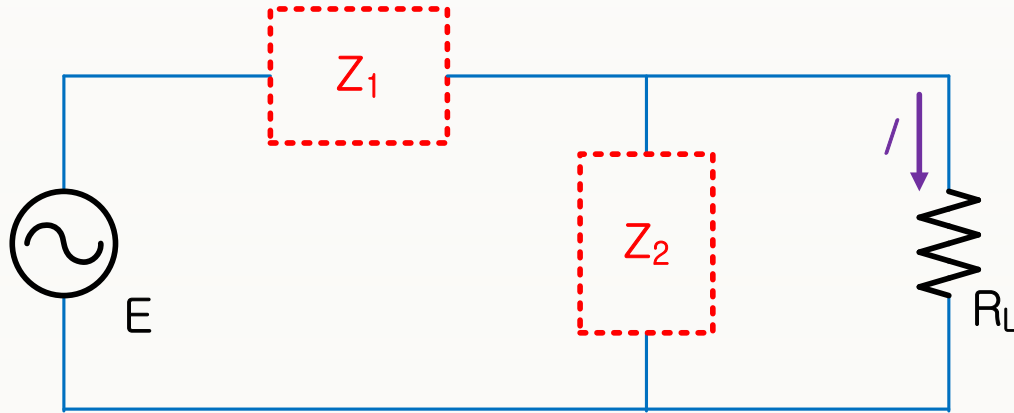
Parallel circuit

$$\frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

Series circuit

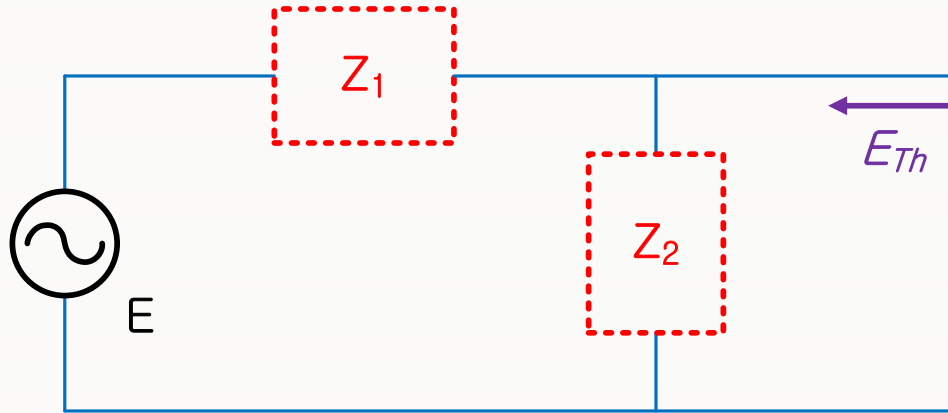
$$Z_1 = R_1 + j\omega L, \quad Z_2 = R_2 + \frac{1}{j\omega C}$$

THEVENIN'S THEOREM

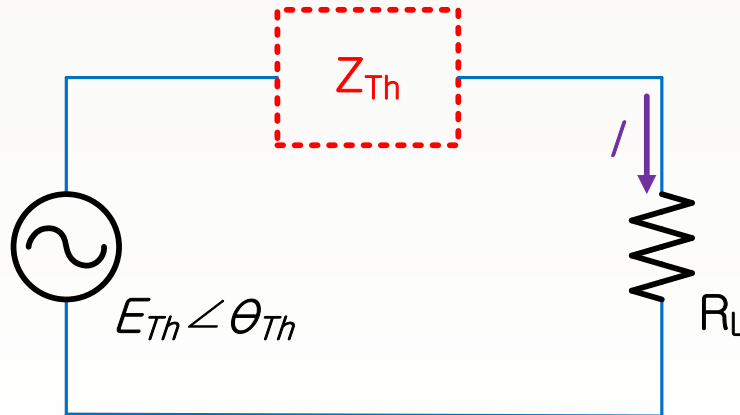


$$Z_{Th} = Z_1 || Z_2$$

THEVENIN'S THEOREM

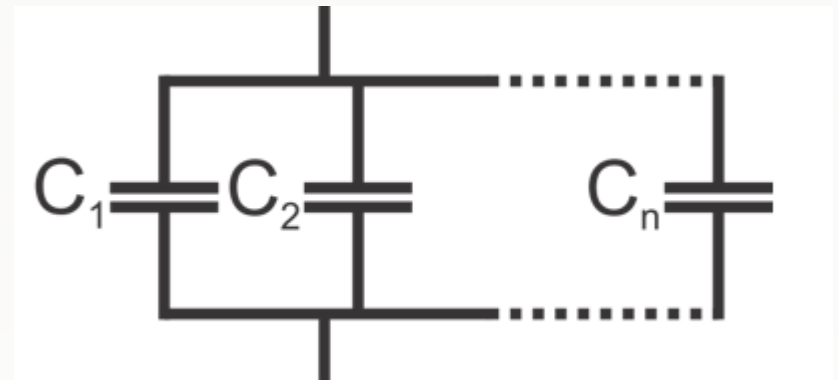
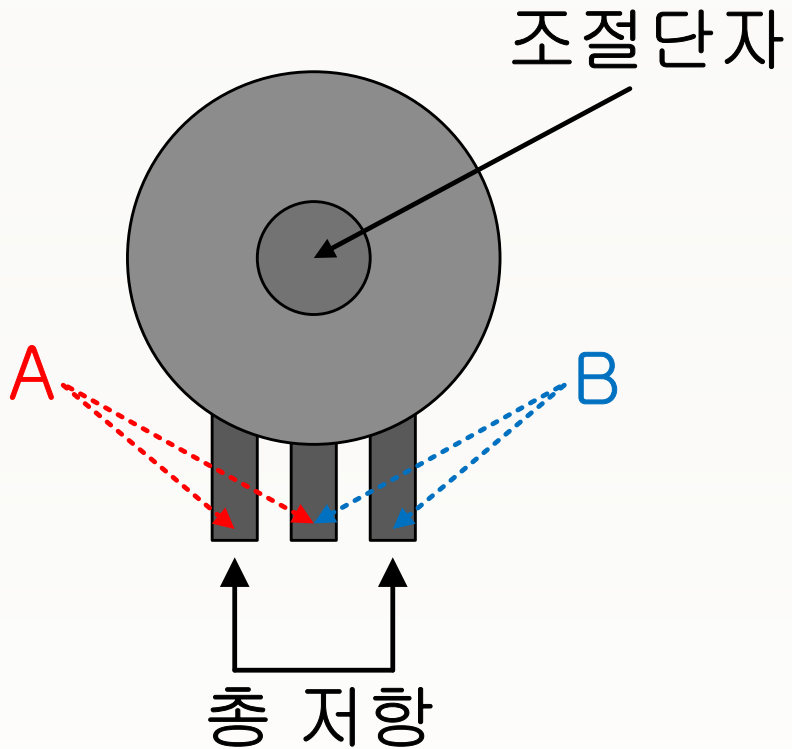


$$E_{Th} = \frac{Z_2}{Z_1 + Z_2} E$$



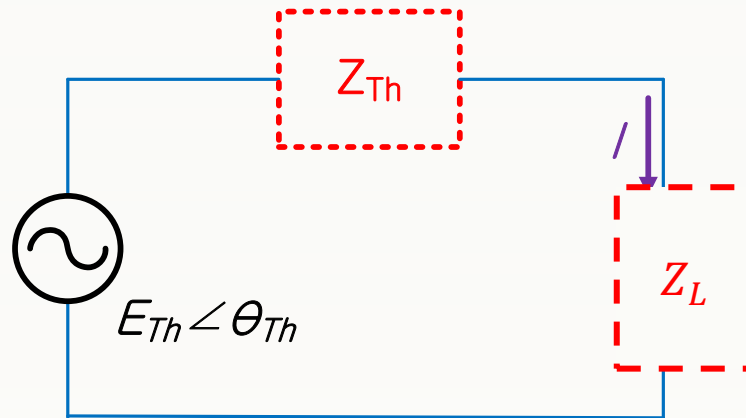
$$I = \frac{E_{Th}}{Z_{Th} + Z_{R_L}}$$

THEVENIN'S THEOREM



$$C_T = C_1 + C_2 + \dots + C_n$$

MAXIMUM POWER TRANSFER



If $Z_{Th} = |Z_{Th}| \angle \theta$ 이면

$Z_L = |Z_{Th}| \angle -\theta$ 일 때 최대 전력이 전달됨

13 주차 실험

▣ AC Ch. 10 : Series-parallel sinusoidal circuits

▣ part3($470\Omega \times 1$, $1k\Omega \times 1$, $10mH \times 1$, $0.02\mu F \times 1$)

- ▣ FIG 10.5 구성 후 $E_s = 4V_{p-p}$ (10kHz) 인가
- ▣ 측정을 통해 Table 10.5, 10.6 완성

▣ AC Ch. 12 : Thevenin's theorem and maximum power transfer

▣ Part1&2($10\Omega \times 1$, $1k\Omega \times 1$, $1.2k\Omega \times 1$, $3.3k\Omega \times 1$, $6.8k\Omega \times 1$, $10mH \times 1$, 가변저항 $\times 1$)

- ▣ FIG 12.4 구성 후 $E_s = 4V_{p-p}$ (10kHz) 인가
- ▣ 측정을 통해 Table 12.1, 12.2 완성

▣ Part3&4($470\Omega \times 1$, 가변저항 $\times 1$, $10mH \times 1$, $0.0047\mu F \times 1$, $0.001\mu F \times 2$, $0.047\mu F \times 1$, $0.1\mu F \times 1$, $1\mu F \times 1$)

- ▣ FIG 12.7 구성 후 $E_s = 4V_{p-p}$ (10kHz) 인가
- ▣ 측정을 통해 Table 12.3($R=470\Omega$ 고정, $V_{ab(p-p)}$), 12.6($C=0.0247\mu F$ 고정, $V_{ab(p-p)}$) 완성